Reduced-Order Modeling of Upper-Level Winds for Fuel-Efficient Flight Routing

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***Abstract*—This study develops and compares data-driven reduced-order models for upper-level wind prediction using openly available ERA5 data. Proper orthogonal decomposition is combined with dynamic mode decomposition and operator inference to model wind-field evolution. The models are evaluated on unseen data using prediction accuracy, computational cost, and a flight route optimization application. Results show that operator inference improves restarted forecast accuracy and reduces model fitting time.**

***Index Terms*—Reduced-order modeling, ERA5, proper orthogonal decomposition, dynamic mode decomposition, operator inference, wind prediction.**

I. Introduction

Atmospheric wind fields are complex, high-dimensional systems exhibiting significant spatial and temporal variations. Their accurate representation and prediction require processing large volumes of meteorological data, making computational efficiency an important consideration. Reduced-order modeling (ROM) provides a data-driven approach to this challenge by identifying dominant structures and representing high-dimensional systems using a smaller number of reduced coordinates.

This study investigates POD-based Dynamic Mode Decomposition (DMD) and Operator Inference (OpInf) for modeling upper-level atmospheric winds using ERA5 reanalysis data at the 250 hPa pressure level. Since the available wind components represent only a single atmospheric pressure level, the resulting system is not dynamically closed. Therefore, non-intrusive, data-driven approaches are adopted to construct and evaluate the reduced-order models.

The models are compared based on their wind-field reconstruction and prediction accuracy, as well as computational efficiency. Additionally, flight-route optimization between Chennai and Singapore is considered as an application to demonstrate the practical utility of the developed reduced-order models.

The source code and implementation are publicly available on GitHub.[1]

II. Literature Review

Numerical Weather Prediction (NWP) is traditionally performed by numerically solving the governing equations of atmospheric motion, together with thermodynamic and continuity equations. These models use observational data assimilation to estimate the initial atmospheric state before advancing the solution in time. The ERA5 reanalysis dataset, developed using the ECMWF Integrated Forecasting System and four-dimensional variational data assimilation, provides an hourly representation of historical atmospheric conditions [1], [2].

Reduced-order modeling has emerged as an effective approach for reducing the computational complexity of high-dimensional physical systems. Proper Orthogonal Decomposition (POD), introduced in the context of coherent structures by Sirovich, identifies dominant spatial modes from a collection of system snapshots. It provides an energy-optimal low-dimensional representation of the available data, allowing the original high-dimensional state to be approximated using a smaller set of modal coefficients [3].

Dynamic Mode Decomposition (DMD), introduced by Schmid, is a data-driven technique that extracts dominant spatial modes and their associated temporal behavior from sequential snapshots. Unlike POD, which primarily identifies energetic structures, DMD provides information about the growth rates and frequencies of the extracted modes. Its application to fluid-flow problems has established it as an important tool for analyzing and predicting complex dynamical behavior [4].

Peherstorfer and Willcox introduced Operator Inference (OpInf) as a non-intrusive approach for constructing reduced-order models directly from simulation data. The method infers reduced operators through least-squares regression without requiring explicit access to the full-order governing operators. It can accommodate linear and polynomial nonlinear reduced dynamics, making it particularly relevant when the original model is available only through data [5]. More generally, projection-based model reduction methods and their computational advantages are discussed by Benner *et al.* [6].

Motivated by these developments, the present study investigates POD-DMD and POD-OpInf for modeling upper-level atmospheric wind fields. Since the available dataset contains only horizontal wind components at a single pressure level, a closed dynamical system is not available for direct intrusive projection. Therefore, the study focuses on non-intrusive data-driven reduced-order modeling and subsequently explores its application to wind-assisted flight routing.

[1] https://github.com/Rakshan-VP/era5-wind-rom-flight-routing

III. Methodology

A. Dataset and State Representation

The ERA5 hourly pressure-level reanalysis dataset, obtained from the Copernicus Climate Data Store (CDS), is used in this study [2], [7]. ERA5 provides hourly atmospheric estimates on a regular latitude–longitude grid, with a horizontal resolution of 0.25° for the reanalysis product [7]. The zonal (u) and meridional (v) wind components at the 250 hPa pressure level are considered.

Hourly data from January 1 to January 31 for each year from 2022 to 2026 are extracted. The geographical domain extends from 0° to 18°N and 78° to 108°E, covering the region between Chennai and Singapore. The resulting grid contains $73 \times 121 = 8833$ spatial points.

At each time instant, the two wind components are concatenated to construct the full-order state vector:

$$\mathbf{x}_k = \begin{bmatrix} u_1 & \cdots & u_{N_g} & v_1 & \cdots & v_{N_g} \end{bmatrix}^T, \tag{1}$$

where $N_g = 8833$ denotes the number of spatial points. The resulting state dimension is

$$n = 2N_g = 17666. \tag{2}$$

Each year contributes 744 hourly snapshots, giving a total of 3720 snapshots. The complete snapshot matrix is therefore

$$\mathbf{X} \in \mathbb{R}^{17666 \times 3720}. \tag{3}$$

When stored in single-precision format, this matrix requires approximately 250.7 MiB of memory.

B. POD Basis Construction and Data Partitioning

Proper Orthogonal Decomposition (POD) is employed to identify the dominant spatial structures in the wind field [3]. The dataset is divided chronologically into a training set containing January data from 2022–2025 and an independent test set containing January 2026. This corresponds to 2976 training snapshots and 744 test snapshots.

The training snapshot matrix is decomposed using singular value decomposition (SVD):

$$\mathbf{X}_{\text{train}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \tag{4}$$

where the columns of $\mathbf{U}$ represent the spatial modes and the diagonal entries of $\mathbf{\Sigma}$ are the singular values.

The cumulative energy captured by the first r modes is defined as

$$E(r) = \frac{\sum_{i=1}^{r} \sigma_i^2}{\sum_{i=1}^{q} \sigma_i^2}, \qquad q = \min(n, m_{\text{train}}), \tag{5}$$

where $m_{\text{train}} = 2976$ is the number of training snapshots.

Three reduced bases are constructed by retaining the minimum number of modes satisfying

$$E(r) \geq \eta, \qquad \eta \in \{0.90, 0.95, 0.99\}. \tag{6}$$

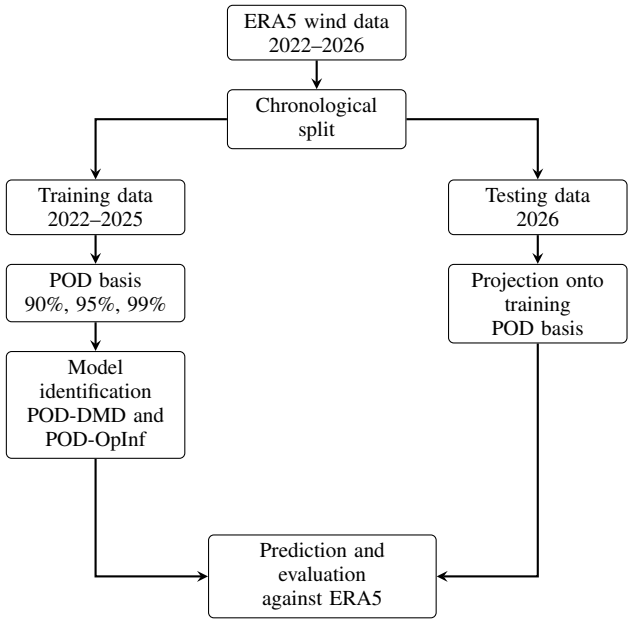


Fig. 1. POD-based reduced-order model development and independent testing workflow.

The corresponding reduced dimensions are $r = 21$, 56, and 269, respectively.

The reduced coordinates are obtained by projecting the full-order state onto the retained POD basis:

$$\mathbf{a}_k = \mathbf{U}_r^T \mathbf{x}_k, \tag{7}$$

where $\mathbf{U}_r \in \mathbb{R}^{n \times r}$ contains the first r POD modes. The corresponding full-order reconstruction is

$$\widehat{\mathbf{x}}_k = \mathbf{U}_r \mathbf{a}_k. \tag{8}$$

The basis is constructed exclusively from the training data. The 2026 snapshots are projected onto this same basis, ensuring that the test data do not influence the POD modes. The projection error provides a reference for evaluating the effect of basis truncation.

The overall ROM development and evaluation procedure is illustrated in Fig.1.

C. Reduced-Order Dynamical Models

The reduced coordinates obtained from POD are used to identify low-dimensional dynamical models. Since only horizontal wind components at a single pressure level are available, the reduced system is treated as a data-driven representation rather than an intrusive projection of a closed governing-equation system.

Two approaches are investigated: POD-based Dynamic Mode Decomposition (POD-DMD) and POD-based Operator Inference (POD-OpInf). DMD identifies a linear evolution operator from time-shifted snapshots [4], whereas OpInf estimates reduced operators directly from data using regression-based operator inference [5]. Both approaches use the same POD basis and reduced dimensions to ensure a consistent comparison.

For DMD, consecutive reduced snapshots are arranged into time-shifted matrices:

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_{m-1} \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} \mathbf{a}_2 & \cdots & \mathbf{a}_m \end{bmatrix}. \tag{9}$$

The reduced evolution operator is estimated using least-squares regression:

$$\mathbf{K}_{\mathrm{DMD}} = \mathbf{B}\mathbf{A}^{\dagger}, \tag{10}$$

where $(\cdot)^{\dagger}$ denotes the Moore–Penrose pseudoinverse. The resulting discrete-time model is

$$\mathbf{a}_{k+1} = \mathbf{K}_{\mathrm{DMD}}\mathbf{a}_k. \tag{11}$$

For OpInf, a regularized linear operator is inferred by minimizing

$$\mathbf{K}_{\mathrm{OpInf}} = \arg\min_{\mathbf{K}} \left[\|\mathbf{B} - \mathbf{K}\mathbf{A}\|_F^2 + \gamma \|\mathbf{K}\|_F^2 \right], \tag{12}$$

where $\|\cdot\|_F$ denotes the Frobenius norm and γ is the regularization parameter. A linear reduced operator with $\gamma = 10$ is used for the reported comparison. The inferred operator advances the reduced state according to

$$\mathbf{a}_{k+1} = \mathbf{K}_{\mathrm{OpInf}}\mathbf{a}_k. \tag{13}$$

The training snapshot pairs are constructed independently within each January period to avoid forming artificial transitions across the gaps between years. The identified operators are subsequently applied to the projected 2026 test data. The predicted reduced coordinates are reconstructed into the full-order wind field using

$$\widehat{\mathbf{x}}_k = \mathbf{U}_r \widehat{\mathbf{a}}_k. \tag{14}$$

D. Evaluation Metrics

The two reduced-order models are evaluated using the held-out January 2026 data. Prediction performance is assessed separately for the zonal and meridional wind components, denoted by ε_u and ε_v. Their mean error is defined as

$$\varepsilon_{uv} = \frac{\varepsilon_u + \varepsilon_v}{2}. \tag{15}$$

The evaluation considers free-running predictions and forecast horizons of 24 h and 72 h. The POD projection error is also used as a reference floor to distinguish errors associated with basis truncation from additional errors introduced by the dynamical models.

The spectral radius of each reduced operator is calculated as

$$\rho(\mathbf{K}) = \max_i |\lambda_i(\mathbf{K})|, \tag{16}$$

where λ_i denotes the eigenvalues of the operator. This quantity is used to examine the stability of the discrete-time reduced dynamics.

Computational efficiency is assessed using model fitting time, prediction time, number of model parameters, and storage requirements. These measures, together with prediction errors, are used to compare POD-DMD and POD-OpInf across the three retained energy levels.

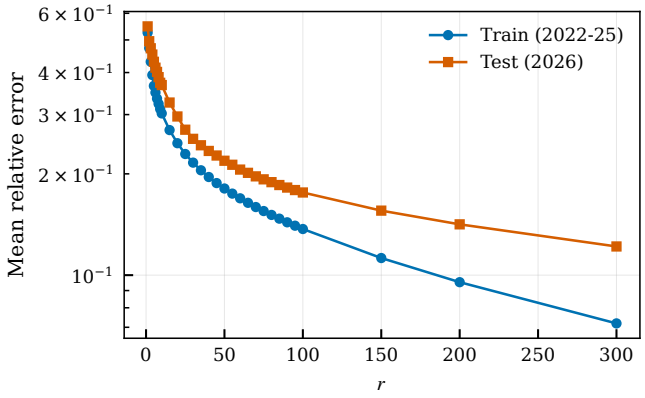


Fig. 2. Wind-field reconstruction error as a function of the number of retained POD modes.

IV. Results and Discussion

A. POD Basis and Reconstruction

The singular-value distribution and cumulative energy captured by the proper orthogonal decomposition (POD) basis are presented together in Fig. 3. The singular values decrease rapidly, indicating that a relatively small number of spatial modes captures a substantial fraction of the variability in the wind field. Based on the cumulative energy criterion, three reduced dimensions were selected: $r = 21$, $r = 56$, and $r = 269$, corresponding to 90%, 95%, and 99% of the training-data energy, respectively.

The reconstruction errors obtained using different numbers of POD modes are shown in Fig. 2. As expected, the reconstruction error decreases as additional modes are retained. However, increasing the reduced dimension also increases the computational cost of identifying and evaluating the reduced-order dynamics. The three selected dimensions therefore provide different levels of spatial representation accuracy for evaluating this trade-off.

The POD basis was constructed exclusively from the training data spanning January 2022 to January 2025. January 2026 was retained as an independent test period. Consequently, the reported test errors evaluate the ability of the reduced-order models to represent wind-field evolution beyond the data used for basis construction and model identification.

B. Reduced-Order Model Selection and Stability

The POD-DMD and POD-OpInf models were evaluated at all three reduced dimensions. For POD-OpInf, linear and quadratic model formulations were considered during validation. The linear formulation with a ridge regularization parameter of $\gamma = 10$ was selected for all three reduced dimensions based on the 2025 validation data. The selected formulation was subsequently fitted using the complete training period from 2022 to 2025.

The validation results used for selecting the POD-OpInf formulation are shown in Fig. 4. The selected linear model provides a consistent basis for comparing the two reduced-order

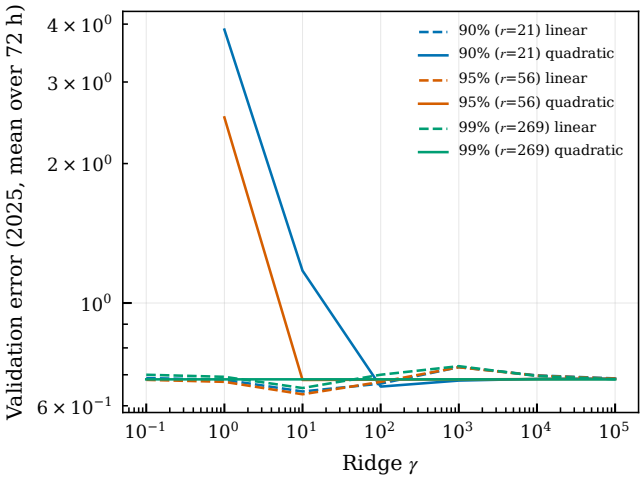


Fig. 4. Validation results for the POD-OpInf model formulations and regularization parameters.

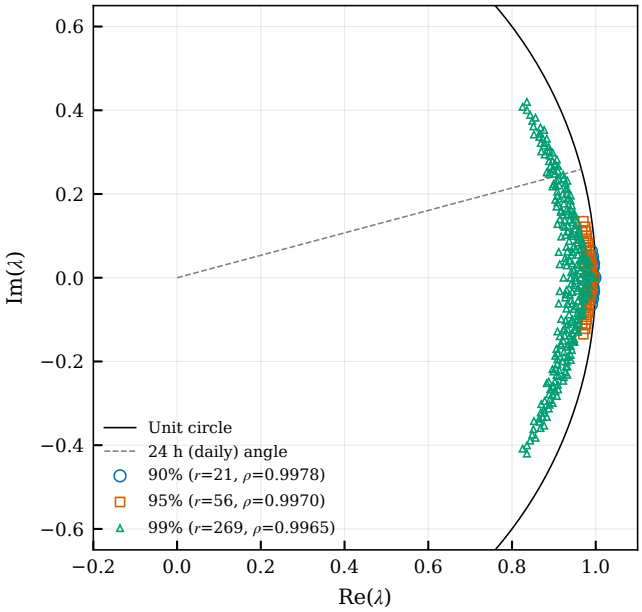


Fig. 5. Eigenvalue distributions of the identified POD-DMD operators for the three reduced dimensions.

modelling approaches without introducing different model structures across the reduced dimensions.

The eigenvalue distributions of the POD-DMD operators are presented in Fig. 5. The spectral radii of the POD-DMD operators are 0.9978, 0.9970, and 0.9965 for $r = 21$, $r = 56$, and $r = 269$, respectively. The corresponding spectral radii of the linear POD-OpInf operators are 0.9944, 0.9937, and 0.9931. All six values are below unity, indicating that the identified linear operators are asymptotically stable in the discrete-time sense. Their proximity to unity also indicates slowly decaying reduced dynamics.

C. Prediction Accuracy

Prediction accuracy was evaluated using the held-out January 2026 data through two procedures: free-run prediction over the test period and restarted forecasts initialized with ERA5 data every 24 h, with lead times up to 72 h. Thus, the lead-time results represent multiple restarted forecasts rather than a continuous 72 h simulation.

Fig. 6 shows the prediction errors, while Fig. 7 presents the relative change between the methods. The reported error is the mean of the normalized zonal and meridional wind errors. Negative percentage changes indicate lower errors for POD-OpInf.

As summarized in Table I, POD-OpInf achieves lower errors for all 24 h and 72 h forecasts across the three POD dimensions. The largest improvement is 7.3% for the 72 h forecast at 90% retained energy. However, free-run performance is mixed: POD-DMD performs marginally better at 90% and 95% energy, while POD-OpInf performs better at 99%.

These results show that improved restarted forecast accuracy does not necessarily translate to better free-run performance. Although increasing the POD dimension reduces projection error, it does not guarantee lower prediction error, since the

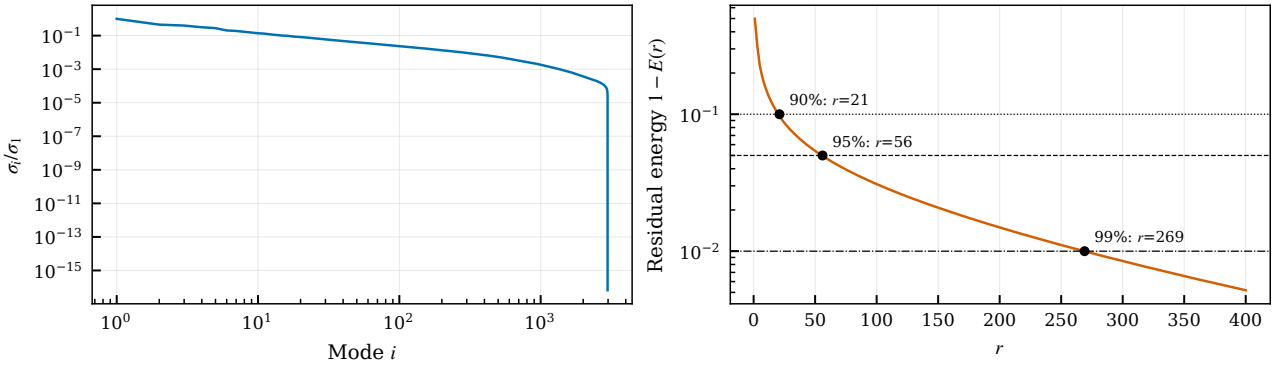


Fig. 3. POD singular-value distribution and residual energy captured by the modes.

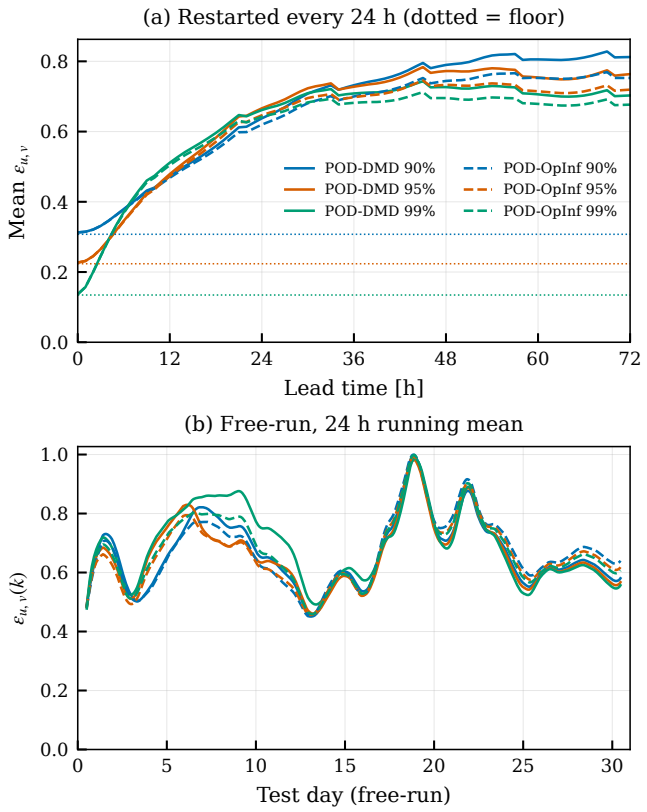


Fig. 6. Comparison of POD-DMD and POD-OpInf prediction errors on the January 2026 test data. The upper panel shows restarted lead-time forecast errors, and the lower panel shows free-run errors.

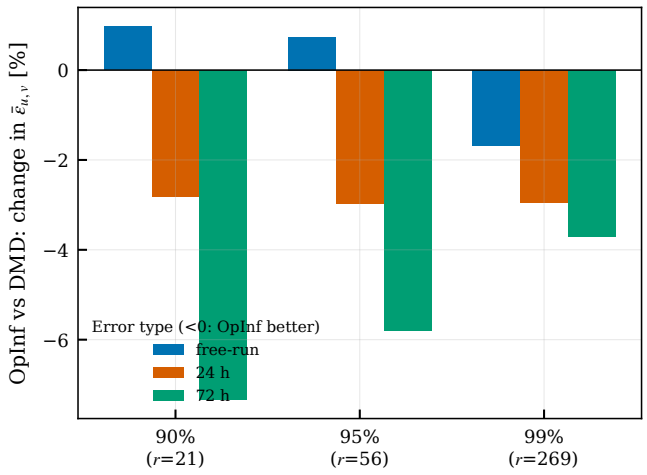


Fig. 7. Percentage change in prediction error of POD-OpInf relative to POD-DMD. Negative values indicate lower errors for POD-OpInf.

latter also depends on the accuracy of the identified reduced dynamics.

TABLE I
MEAN NORMALIZED PREDICTION ERRORS ON THE JANUARY 2026 TEST DATA.

Energy	Evaluation	DMD	OpInf
90%	Free-run	0.6629	0.6694
	24 h	0.6377	0.6198
	72 h	0.8122	0.7526
95%	Free-run	0.6576	0.6625
	24 h	0.6662	0.6464
	72 h	0.7636	0.7194
99%	Free-run	0.6864	0.6749
	24 h	0.6613	0.6418
	72 h	0.7028	0.6767

D. Computational Performance

The computational performance of the two approaches was assessed using model fitting time, prediction time, and estimated storage requirements. The results are summarized in Table II. POD-OpInf requires less fitting time than POD-DMD for all three reduced dimensions. The reduction in fitting time is approximately 74.3% for $r = 21$, 68.7% for $r = 56$, and 89.0% for $r = 269$. The difference becomes particularly significant at the largest reduced dimension, where the POD-OpInf fitting time is 0.0139 s compared with 0.1262 s for POD-DMD.

The prediction times, however, are comparable and do not show a consistent advantage for either method. POD-OpInf is marginally slower at $r = 21$ and $r = 56$, while it is faster at $r = 269$. Therefore, the primary computational advantage observed for POD-OpInf in this implementation is its reduced model fitting time rather than a consistent improvement in prediction speed.

The estimated storage requirements are identical for the two approaches at each reduced dimension. Storage increases from 2.97 MiB at $r = 21$ to 36.94 MiB at $r = 269$, reflecting the increased size of the retained POD basis and reduced-order model representation. These results show that POD-OpInf can reduce model identification cost without introducing an increase in the estimated storage requirement.

E. Discussion

POD-DMD achieves slightly lower free-run errors at 90% and 95% retained energy, while POD-OpInf consistently improves the restarted 24 h and 72 h forecasts, with a maximum error reduction of 7.3%. POD-OpInf also reduces fitting time by 68.7–89.0%, while prediction times and storage requirements remain comparable. These results indicate that neither method is universally superior; performance depends on the reduced dimension and prediction setting. Overall, POD-OpInf provides a favourable balance of forecast accuracy and fitting efficiency for the evaluated forecasts.

F. Flight Route Application

A time-dependent route optimization was performed for the Chennai–Singapore corridor using a constant airspeed of 250

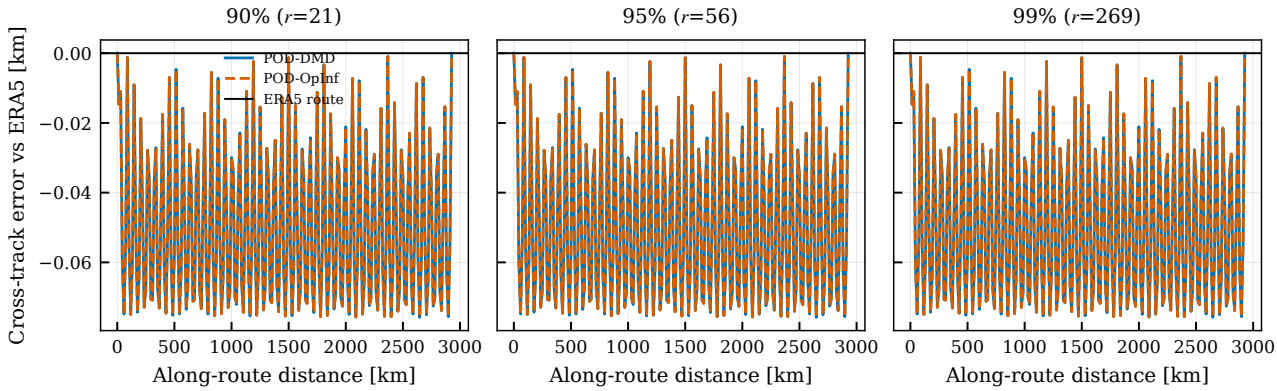


Fig. 8. Cross-track error of POD-DMD and POD-OpInf optimized routes relative to the ERA5 reference route for a 13 January 2026 departure at 00:00 UTC. The panels show results for 90%, 95%, and 99% retained POD energy.

TABLE II
STABILITY AND COMPUTATIONAL COST OF POD-DMD AND POD-OPINF.

Energy	Metric	DMD	OpInf
90%	Spectral radius	0.9978	0.9944
	Fit time (s)	0.0056	0.0014
	Prediction time (s)	0.0766	0.0773
	Storage (MiB)	2.97	2.97
95%	Spectral radius	0.9970	0.9937
	Fit time (s)	0.0119	0.0037
	Prediction time (s)	0.0704	0.0834
	Storage (MiB)	7.71	7.71
99%	Spectral radius	0.9965	0.9931
	Fit time (s)	0.1262	0.0139
	Prediction time (s)	0.1215	0.1133
	Storage (MiB)	36.94	36.94

m/s. Routes generated using POD-DMD and POD-OpInf wind predictions were compared against the ERA5 reference route. The signed cross-track error was calculated along the route for a 13 January 2026 departure at 00:00 UTC. Fig. 8 compares the deviations for the three POD energy levels.

V. CONCLUSION

This study demonstrates that useful reduced-order models can be developed from openly available ERA5 wind data without requiring access to the underlying governing equations. Both POD-DMD and POD-OpInf capture the dominant wind-field behaviour, with POD-OpInf offering improved restarted forecast accuracy and lower fitting time. The results highlight the potential of data-driven ROMs for atmospheric applications, while also revealing limitations in long-term free-run prediction.

The main observations are:

- POD provides substantial dimensionality reduction while retaining the dominant wind-field energy.
- POD-OpInf improves all evaluated restarted forecasts compared with POD-DMD.
- Higher retained energy does not necessarily guarantee lower prediction error.
- ROM-based wind predictions can support route optimization, as demonstrated by the cross-track error comparison.

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